

DATA INFRASTRUCTURE DOCUMENTATION

Database Reference Sample

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Technical Writing Portfolio

Portfolio Sample

This document is an original portfolio sample created to demonstrate techniques used in database and data-infrastructure documentation. Names, identifiers, descriptions, and other details have been fictionalized.

Introduction

This chapter contains a detailed description of every table in the Admin database, in alphabetical order. Each table description includes the following information:

- table name and brief description of table
- list of modules that use the table
- list of tables that depend on the table
- descriptions of triggers, check constraints, foreign keys, and indexes defined on the table
- descriptions of every field in the table

Each field description includes the following information:

- field name
- data type and size (see below)
- comments—brief field description
- values—rules that constrain field values (preceded by R:) and default values for the field (preceded by D:)
- null—indicates whether a Null value is allowed in the field

The data type associated with each field depends on the relational database (RDB) system your deployment uses. For example, the `is_default` field in `ACCESS_REGION_REF` is a `Number(1)` field in an Oracle database and a `bit` field in a Sybase database.

Data Types

Every field description in this document contains a single documented data type. Table 2, “Field Types,” lists the valid documented data types and their corresponding RDB data types for Oracle and Sybase.

Table 2 Field Types

Documented Type	Oracle Type	Sybase Type	Minimum Value	Maximum Value
bit	Number(1,0)	bit (1 byte)	0	1
char[N]	char(N)	char[N]	N characters	N characters
Fixed-length string, padded with blanks if smaller than N. N cannot exceed 255.				
datetime	date	datetime (8 bytes)	varies	varies
For Oracle, stores date/time between January 1, 4712 B.C. and December 31, 4712. For Sybase, stores date/time between January 1, 1753 and December 31, 9999.				
image	blob	image (16 bytes)	—	—
For Oracle, stores variable-length binary data up to 2 GB. For Sybase, stores a pointer to a memory space allocated in multiples of 2 KB that cannot exceed 2 GB.				
int	Number(10,0)	int (4 bytes)	-2,147,483,648 (-2 ³¹)	2,147,483,647 (2 ³¹ - 1)
numeric	Number(18,0)	numeric (10) (8 bytes)	-999,999,999,999,999,999	999,999,999,999,999,999
smalldt	date	smalldatetime (4 bytes)	varies	varies
For Oracle, stores date/time between January 1, 4712 (B.C.) and December 31, 4712. For Sybase, stores date/time between January 1, 1900 and June 6, 2079.				
smallint	Number(6,0)	smallint (2 bytes)	-32,768 (-2 ¹⁵)	32,767 (2 ¹⁵ - 1)
text	varchar2(4000)	text (16 bytes)	—	—
For Oracle, stores variable-length character data up to 4000 bytes. For Sybase, stores a pointer to a memory space allocated in multiples of 2 KB that cannot exceed 2 GB.				
tinyint	Number(3,0)	tinyint (1 byte)	0	255
varchar[N]	varchar2(N)	varchar[N]	0 characters	N characters
Variable-length string. N cannot exceed 255.				

Table 2 also includes minimum and maximum values for each documented type. These are not necessarily the minimum and maximum supported by your database; rather, they are the minimum and maximum values supported by Kenan BP for fields of this type. Therefore, you should not enter field values that exceed the minimum or maximum values documented here, even if the corresponding RDB data type allows those values.

The value in parentheses is the physical size of the field. The physical size of the field is the number of bytes needed to store a field.

In the example below, the character limit for short_display is 5 characters and the physical size of the field is 15 bytes.

Table 3 Sample Row

<i>Field Name</i>	<i>Data Type</i>	<i>Comments</i>	<i>Values</i>	<i>Null</i>
short_display	varchar (15) 5	Short text description for ABI source.		Null

Table: EXTERNAL_ID_TYPE_REF

Configurable. Defines external ID types. Admin table. Read-only copy exists on Catalog database.

Used by the following modules

Customer Configuration

Dependents

CUSTOMER_ID_ACCT_MAP, CUSTOMER_ID_EQUIP_MAP, CUSTOMER_ID_EQUIP_MAP_KEY, CUSTOMER_ID_EQUIP_MAP_VIEW, CUSTOMER_ID_GROUP_MAP, EXTERNAL_ID_ACCT_MAP, EXTERNAL_ID_EQUIP_MAP, EXTERNAL_ID_EQUIP_MAP_KEY, EXTERNAL_ID_EQUIP_MAP_VIEW,

Triggers

Only one row can have `is_default = 1`.

`is_default` cannot be updated to 0.

`is_viewable` cannot be updated to 0.

Rows with `is_default` set to 1 cannot have `uniqueness_type` set to 0.

Indexes

primary-key on `external_id_type`

Check Constraints

`is_default != 1 or is_viewable = 1`

Field Name	Data Type		Comments	Values	Null
external_id_type	smallint		External ID type ID associated with source.	(R: != 0)	
uniqueness_type	tinyint		How are <code>external_id</code> fields of this type mapped to internal IDs? 0 = one external ID is mapped to many internal IDs (one-to-many). 1 = unique; one external ID is mapped to one internal ID (one-to-one). 2 = more than one external ID is mapped to one internal ID (many-to-one). 3 = reuse default external ID for subscriber disconnection.	(R: 0, 1, 2, 3)	

Field Name	Data Type		Comments	Values	Null
is_default	bit		Is this value default? (1 = yes) Only one row can be default. When a module needs an unspecified row from this table it uses the default row. See the <i>Module Configuration Guide</i> and the <i>Technical Reference</i> for more information.	(D: 0)	
is_internal	bit		If 1, Kenan/BP modules require this record; do not delete or modify.	(D: 0)	
is_viewable	bit		Is this value viewable through the GUI? (1 = yes) This field must be set to 1 when <code>is_default = 1</code> .	(D: 1)	
config_tag_id	varchar	32	This column is client specific and not used by Kenan processes.	(D: Kenan FX 3.0)	Null

Table EXTERNAL_ID_TYPE_VALUES

Configurable. Describes external ID types defined in `EXTERNAL_ID_TYPE_REF`. Run the `replicate.ksh` script after modifying this table if the database is in a multiserver environment. Admin table. Read-only copy exists on Customer database.

Indexes

primary-key on external_id_type, language_code

Foreign Keys

(external_id_type) must exist in `EXTERNAL_ID_TYPE_REF` (external_id_type) (language_code) must exist in `LANGUAGE_CODE_REF` (language_code)

Field Name	Data Type		Comments	Values	Null
external_id_type	smallint		External ID type ID, from <code>EXTERNAL_ID_TYPE_REF</code> .		
language_code	smallint		Language code, from <code>LANGUAGE_CODE_REF</code> .		
short_display	varchar (15)	5	Abbreviated version of the display value.		Null
display_value	varchar (240)	80	Text description to display on screen.		
config_tag_id	varchar	32	This column is client specific and not used by Kenan processes.	(D: Kenan FX 3.0)	Null

Data Model Notes

EXTERNAL_ID_TYPE_REF defines the recognized types of external identifiers used by the application. Each identifier type is represented by a unique reference value.

EXTERNAL_ID_TYPE_VALUES contains the valid values associated with those identifier types.

The relationship between the tables allows the database to maintain a controlled set of identifier types and their corresponding values. This structure separates the definition of a reference category from the values that belong to that category.

